



Self Driving Challenge

Closed Category introduction



February 12, 2026



The Track

Track

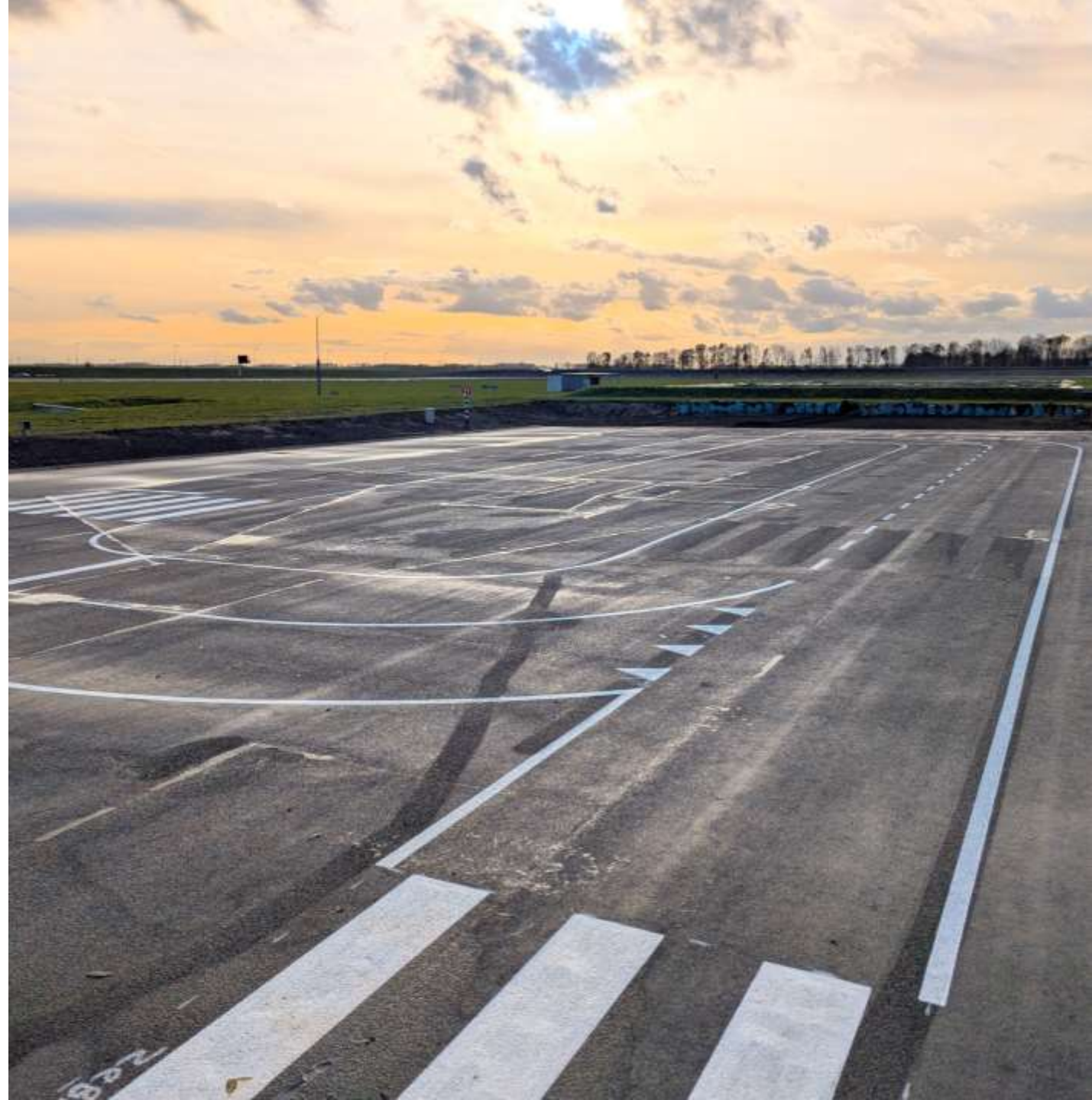




Road markings

- Lane markings
- Intersections
- Pedestrian crossings
- Give-way road-markings

- Follow your lane
- Change lane as needed
- Navigate the course



Traffic lights

- Traffic lights are remote controlled, manually operated
- Operated by teams during testing
- Operated by RDW during final race



Traffic signs

- Speed limits
- Entry/directional restrictions
- Position and orientation could vary slightly every testing day and final race day



VRU robot

- Simulates a pedestrian crossing the street
- Can be controlled remotely
- Battery-powered, remember to check and/or ask for recharging!
- Operated by teams during testing
- Operated by RDW during final race



Soft target (car)

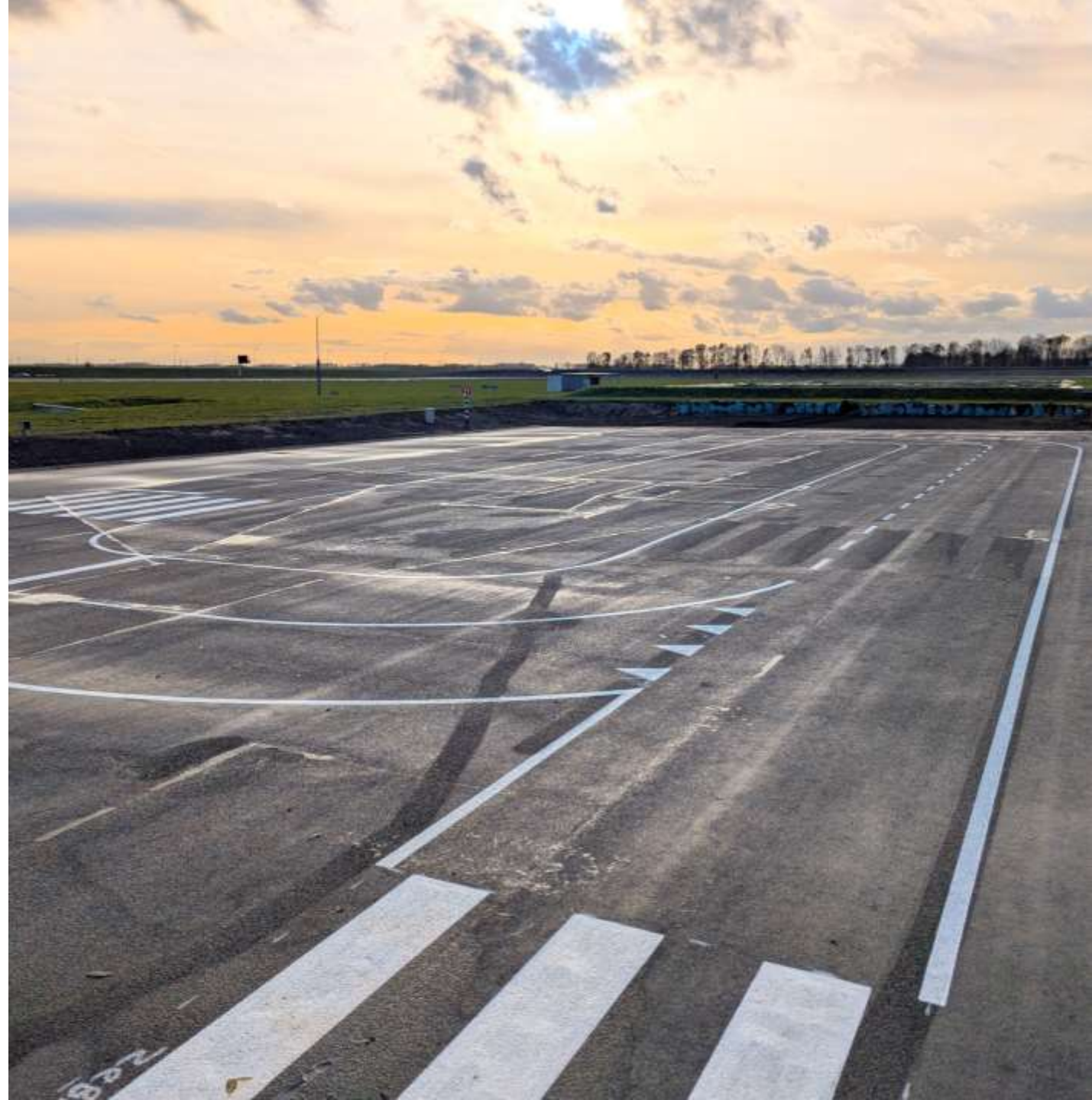
- Fixed position (not moving)
- To be overtaken





Route planning

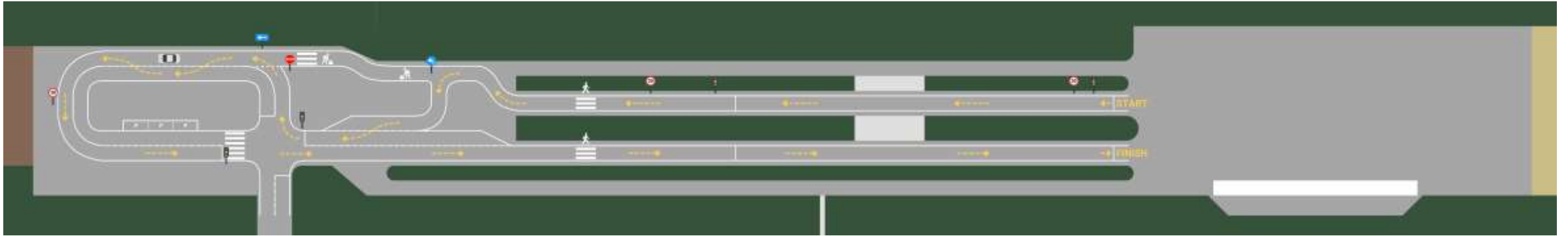
- The final obstacle setup and its route will only be disclosed the day of the finale
- You need to handle navigation and dynamic object encounters
- During test days you can place traffic signs, lights, etc. where you want to develop your solution



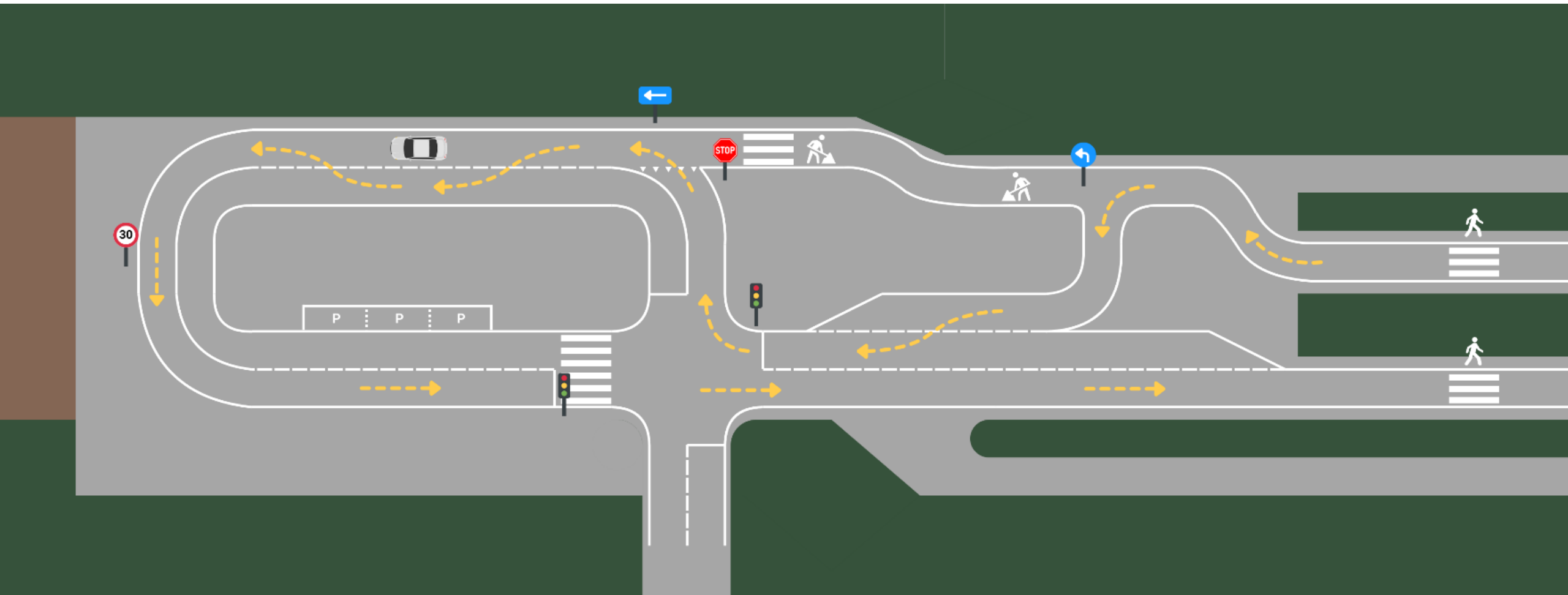
Track



Track



Track

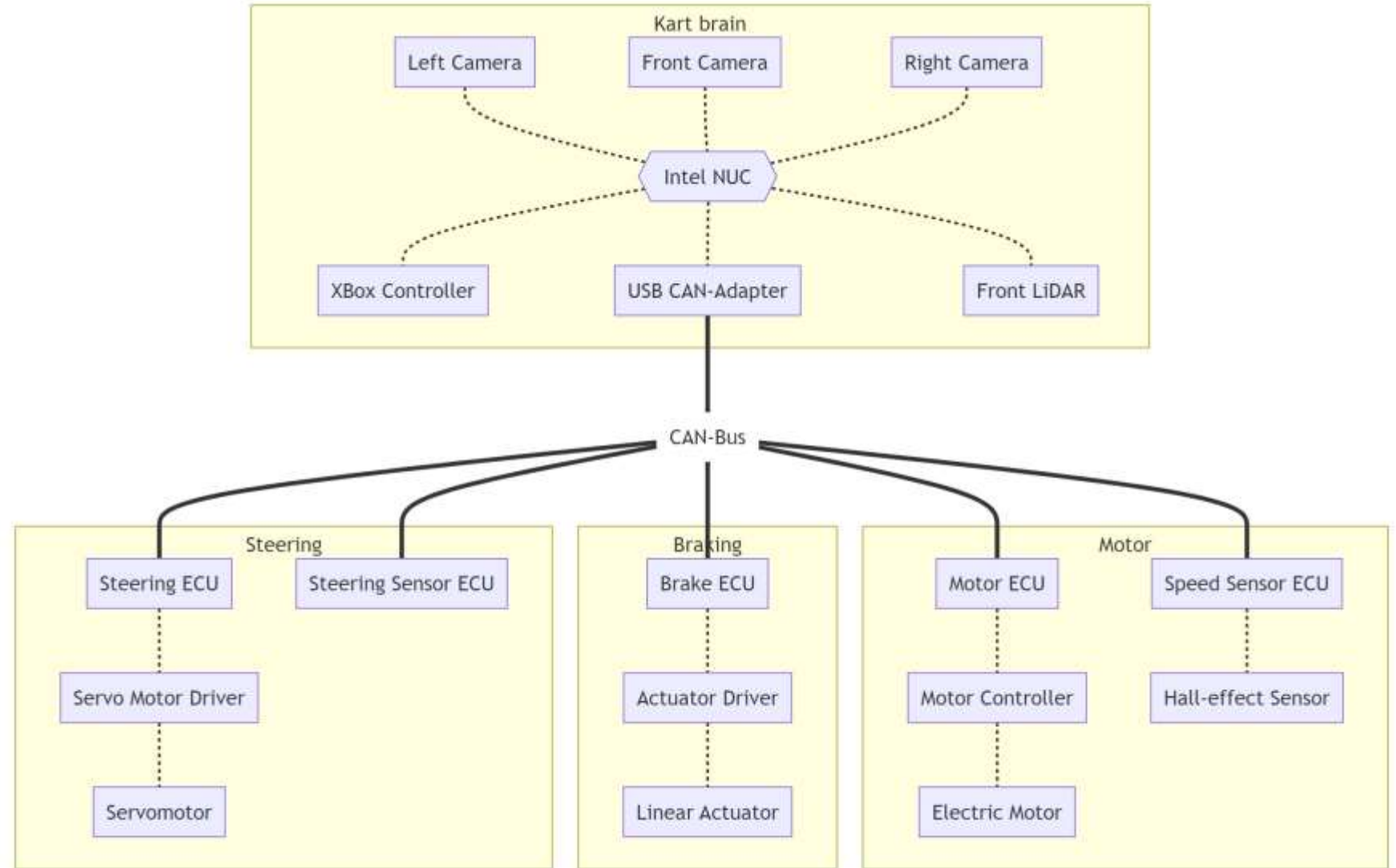




The Kart

The kart

- Sensor setup
 - Cameras
 - LiDAR
 - Speed sensor
- Controlling the kart
 - CAN bus
 - USB CAN interface



A solid line depicts the CAN-bus where a dotted line depicts other connections and protocols.

Starting the kart

1. Connect the batteries
2. Computer + brake power available
3. On-board emergency stop
4. Remote emergency stop
5. Kart system power available
6. Speed (limit) selection
 - Start with the lowest setting, ask permission for higher speeds.
7. Check for errors/problems
8. Ready!



Controlling the kart yourself

- You can control the kart yourself
 - Useful for e.g. resetting to the start
 - Only with a controller
 - Wearing a helmet is mandatory
 - Remote stop button mandatory
- Controls
 - Long press A to activate
 - Tripple press A to enable / disable
 - Left trigger to brake
 - Right trigger to accelerate
 - Left joystick to steer
 - X, Y, B for forward, neutral, reverse



The computer

- Every team gets their own NUC
 - Provided on loan
 - Along with a controller, camera and LiDAR for testing
- Switching the NUC on the kart
 - Can be done in ~20 seconds



Networking

- Wifi is available trackside and on kart
- Trackside:
 - **SDC AP 1**
 - **SDC AP 2**
- On kart
 - **SDC Kart 1**
 - **SDC Kart 2**
- Password is **SDC_2025**
- *Tip: you could set up a VPN to communicate with the NUC remotely*
- *Tip: Check that your NUC connects to the right AP! And not to the AP on the trackside or other kart, if applicable.*





Testing Days

Schedule



EDITION 2026 SCHEDULE

Legend

X Testing days in Lelystad

X Kick-off & final race

February				
Mon	Tue	Wed	Thu	Fri
2	3	4	5	6
9	10	11	12	13
16	17	18	19	20
23	24	25	26	27

March				
Mon	Tue	Wed	Thu	Fri
2	3	4	5	6
9	10	11	12	13
16	17	18	19	20
23	24	25	26	27
30	31			

April				
Mon	Tue	Wed	Thu	Fri
		1	2	3
6	7	8	9	10
13	14	15	16	17
20	21	22	23	24
27	28	29	30	

May				
Mon	Tue	Wed	Thu	Fri
				1
4	5	6	7	8
11	12	13	14	15
18	19	20	21	22
25	26	27	28	29

June				
Mon	Tue	Wed	Thu	Fri
1	2	3	4	5
8	9	10	11	12
15	16	17	18	19
22	23	24	25	26
27				

Sector 9

- Portable radio's available at the reception
- Ask for permission at the stopsign when crossing the oval track; *"Self Driving Challenge for the tower..."*
 - "...i want to cross from the outside to sector 9."
 - "...i want to cross from inside to the outside."
- **Always** wait for **your own** permission, even if the barrier is already open or another car is already crossing!
- Park at the designated parking area near Sector 9A.
- Always drive carefully and give way!





Questions?



Time	Activity
11:00	Arrival & registration with lunch
12:00	Start official program
13:15	Group photo (outside) + short break
13:30	Explanation closed category (Hall 5) + Round 1: team photos / team videos and Kombaan track lap
14:15	Explanation open category (Hall 5) + Round 2: team photos / team videos and Kombaan track lap
15:00	Networking drinks
16:00	End of program